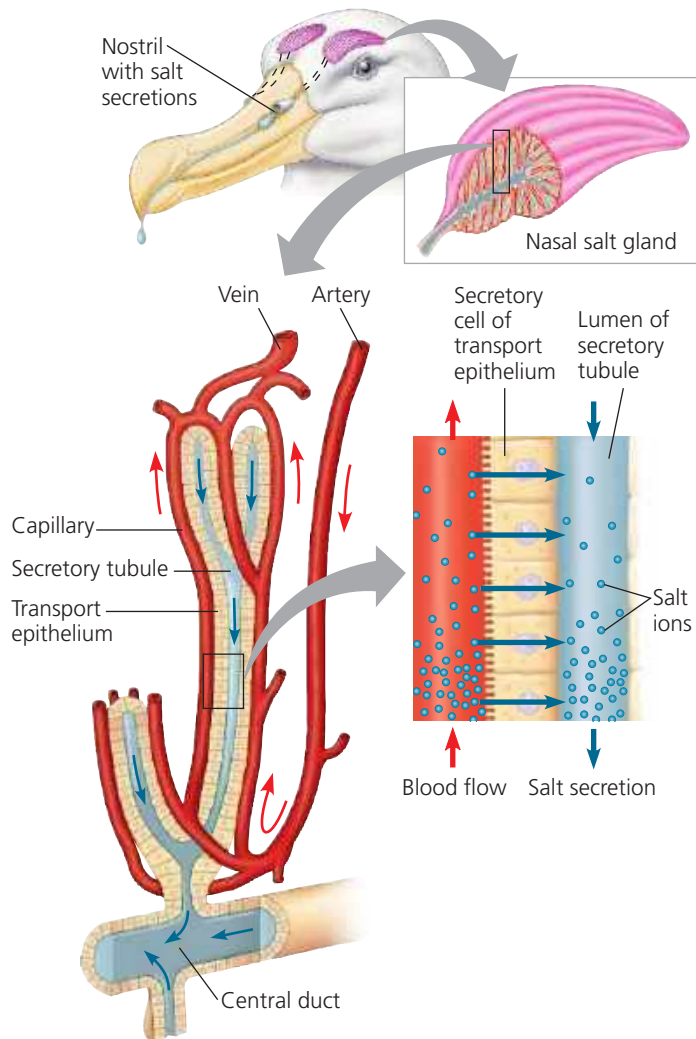


▼ **Figure 44.5 Salt secretion in the nasal glands of a marine bird.** A transport epithelium moves salt from the blood into secretory tubules, which drain into central ducts leading to the nostrils.



CONCEPT CHECK 44.1

1. The movement of salt from the surrounding water to the blood of a freshwater fish requires the expenditure of energy in the form of ATP. Why?
2. Why aren't any freshwater animals osmoconformers?
3. **WHAT IF?** Researchers found that a camel in the sun required much more water when its fur was shaved off, although its body temperature was the same. What can you conclude about the relationship between osmoregulation and the insulation provided by fur?

For suggested answers, see Appendix A.

CONCEPT 44.2

An animal's nitrogenous wastes reflect its phylogeny and habitat

In regulating and safeguarding their internal fluids, animals must deal with **ammonia**, a toxic metabolite produced by the dismantling of *nitrogenous* (nitrogen-containing) molecules,

chiefly proteins and nucleic acids. Several mechanisms have evolved for ridding the body of ammonia and other metabolic waste products, a process called **excretion**. Because most metabolic wastes must be dissolved in water to be excreted from the body, the type and quantity of an animal's waste products may have a large impact on its water balance.

Forms of Nitrogenous Waste

Although some animal species excrete ammonia directly, others excrete alternative forms of nitrogenous waste, either urea or uric acid (**Figure 44.6**). These different forms vary significantly in their toxicity, their solubility, and the energy costs of producing them.

Ammonia

Ammonia is very toxic, in part because its ion, ammonium (NH_4^+), can interfere with oxidative phosphorylation. Because ammonia can be tolerated only at very low concentrations, animals that excrete ammonia need access to lots of water. Therefore, ammonia excretion is most common in aquatic species. The highly soluble ammonia molecules, which interconvert between NH_3 and NH_4^+ , easily pass through membranes and are readily lost by diffusion to the surrounding water. In many invertebrates, ammonia release occurs across the whole body surface.

▼ **Figure 44.6 Variations in forms of nitrogenous waste among animal species.**

